Code-Layout, Readability and Reusability

Date	1 July 2026
Team ID	SWTID-2026-8096
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	5 Marks

Code Layout Checklist

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation are maintained throughout the code.	Yes	Improves readability and code consistency.
2	Proper File Structure	Project files and folders are organized logically	Yes	Separate folders for templates, static files, model, and dataset.
3	Meaningful Variable Names	Variables are named according to their purpose.	Yes	Makes the code easy to understand and maintain.
4	Function / Method Names	Functions use descriptive names.	Yes	Functions clearly indicate their operations.
5	Code Comments	Important sections include explanatory comments.	Yes	Comments help developers understand the implementation.
6	Modular Design	Code is divided into reusable modules and functions.	Yes	Separate modules for prediction, preprocessing, and web application.
7	No Redundant Code	Duplicate and unused code has been removed.	Yes	Reduces complexity and improves maintainability.
8	Error Handling	Invalid inputs and runtime errors are handled gracefully.	Yes	Input validation and exception handling improve reliability.

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Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	Prediction Module	Python, Scikit-learn	Flood prediction using the trained Random Forest model	High
2	Data Preprocessing Module	Python, Pandas, NumPy	Dataset cleaning and preprocessing	High
3	Flask Application (app.py)	Python, Flask	Handles routing, user requests, and predictions	High
4	HTML Templates	HTML5, CSS3	Home page, prediction page, and result pages	Medium
5	CSS Styling	CSS3	Shared styling across all web pages	High

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project with separate folders for templates, static files, dataset, and model.
Readability	5	Meaningful variable names, proper indentation, and descriptive functions make the code easy to understand.
Reusability	5	Modular design allows prediction logic, preprocessing, and UI components to be reused.
Documentation / Comments	4	Important sections are documented with comments for easier maintenance.
Overall Score	4.8/5	The code is clean, modular, maintainable, and suitable for future enhancements.